Day 10 Problem Set

Geometry Summer Credit | Right-Triangle Similarity and Weekly Review

Name:	Date:	Period:

Goal

Practice introductory right-triangle similarity and review quadrilaterals, similarity, scale factor, and proportions.

Support Practice

1.	Solve: $4/5 = x/30$.

2.	Solve: $9/x = 3/8$.

3.	Find the midpoint of (2, -6) and (10, 4).

4.	A rectangle has diagonals $2x + 5$ and 31. Find x.

5.	A dilation by scale factor 3 maps (4, -2) to what point?

Core Practice: Right-Triangle Similarity

6.	An altitude to the hypotenuse splits the hypotenuse into segments 3 and 12. Use h^2 = pq to find h.

7.	An altitude to the hypotenuse splits the hypotenuse into segments 8 and 18. Find h.

8.	A right triangle has hypotenuse 20. A leg projects to a segment of length 5 on the hypotenuse. Use leg^2 = cp to find the leg.

9.	A right triangle has hypotenuse 27. A leg projects to a segment of length 12 on the hypotenuse. Find the leg.

10.	In a right-triangle altitude diagram, why are the smaller triangles similar to the original triangle?

11.	Error analysis: A student uses h = p + q instead of h^2 = pq. Explain the mistake.

Day 10 Problem Set

Continued

Core Practice: Weekly Review

12.	A parallelogram has consecutive angles 3x and x + 20. Find x and both angles.

13.	A rhombus has perimeter 84. Find each side length.

14.	A rectangle has diagonals 5x - 7 and 38. Find x.

15.	Triangle ABC is similar to triangle DEF. AB = 9, DE = 24, AC = 15. Find DF.

16.	A 6-foot person casts an 8-foot shadow. A building casts a 44-foot shadow. Find the building height.

17.	A map scale is 1 inch = 18 miles. Two cities are 4.5 inches apart. Find the real distance.

18.	A triangle has vertices (0,0), (5,0), (0,7). A similar triangle has vertices (0,0), (15,0), (0,21). Find the scale factor.

19.	Find the distance between A(-1, 2) and B(7, 8).

20.	Can 7, 11, and 19 form a triangle? Explain.

21.	Challenge: A rectangle is scaled by factor 4. What happens to perimeter and area?